Babu Banarasi Das University



NO SQL PROJECT ON

Student Record Management System

Submitted to: Mr. Harendra Singh

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OBJECTIVES:

- To understand the concept of NoSQL databases.
- To create a database and collection in MongoDB.
- To insert and manage 10 student records.
- To perform CRUD operations — Create, Read, Update, and Delete.
- To use comparison operators such as `$gt, $gte, $lt, $lte, $ne,` and `$in`.
- To perform sorting and filtering of student records.
- To perform search and retrieval operations based on specific conditions.
- To understand and implement AND (`$and`) and OR (`$or`) operations.
- To understand how MongoDB can be used to efficiently manage and retrieve student information.

By Aanya Gupta

1. CREATE DATABASE

MongoDB Query:

- *use AcademicDB*



2. CREATE COLLECTION

MongoDB Query:

- *db.createCollection("students")*

```
> db.createCollection("students")
< { ok: 1 }
```

3. INSERT 10 STUDENTS

MongoDB Query:

- *db.students.insertMany([ ])*

```
db.students.insertMany([
{roll:1, name:"Aanya Gupta", age:19,marks:88,city:"Lucknow"},
{roll:2, name:"Riya Sharma", age:18,marks:76,city:"Delhi"},
{roll:3, name:"Rahul Verma", age:19,marks:92,city:"Kanpur"},
{roll:4, name:"Aman Singh", age:18,marks:69,city:"Delhi"},
{roll:5, name:"Neha Yadav", age:19,marks:81,city:"Lucknow"},
{roll:6, name:"Karan Mehta", age:20,marks:85,city:"Kanpur"},
{roll:7, name:"Pooja Mishra", age:19,marks:73,city:"Lucknow"},
{roll:8, name:"Arjun Nair", age:20,marks:90,city:"Delhi"},
{roll:9, name:"Sneha Rao", age:18,marks:58,city:"Kanpur"},
{roll:10, name:"Vivek Joshi", age:20,marks:55,city:"Delhi"}
])
```


CONNECTIONS (1)
Search connections
Aanya
admin
config
local
> db.students.insertMany([
{roll:1, name:"Aanya Gupta", age:19,marks:88,city:"Lucknow"},
{roll:2, name:"Riya Sharma", age:18,marks:76,city:"Delhi"},
{roll:3, name:"Rahul Verma", age:19,marks:92,city:"Kanpur"},
{roll:4, name:"Aman Singh", age:18,marks:69,city:"Delhi"},
{roll:5, name:"Neha Yadav", age:19,marks:81,city:"Lucknow"},
{roll:6, name:"Karan Mehta", age:20,marks:85,city:"Kanpur"},
{roll:7, name:"Pooja Mishra", age:19,marks:73,city:"Lucknow"},
{roll:8, name:"Arjun Nair", age:20,marks:90,city:"Delhi"},
{roll:9, name:"Sneha Rao", age:18,marks:58,city:"Kanpur"},
{roll:10, name:"Vivek Joshi", age:20,marks:55,city:"Delhi"}
])

4. DISPLAY ALL STUDENTS

MongoDB Query:

- *db.students.find()*

```
> db.students.find()
< {
    _id: ObjectId('6b1d0c4e7a92f3b5c8e1a000'),
    roll: 1,
    name: 'Aanya Gupta',
    age: 19,
    marks: 88,
    city: 'Lucknow'
  }
  {
    _id: ObjectId('6b1d0c4e7a92f3b5c8e1a001'),
    roll: 2,
    name: 'Riya Sharma',
    age: 18,
    marks: 76,
    city: 'Delhi'
  }
  {
    _id: ObjectId('6b1d0c4e7a92f3b5c8e1a002'),
    roll: 3,
    name: 'Rahul Verma',
    age: 19,
    marks: 92,
    city: 'Kanpur'
  }
```

5. EQUAL TO — $eq

MongoDB Query:

- *db.students.find({marks: {$eq: 76}})*

```
> db.students.find({marks: {$eq: 76}})
< {
    _id: ObjectId('6b1d0c4e7a92f3b5c8e1a001'),
    roll: 2,
    name: 'Riya Sharma',
    age: 18,
    marks: 76,
    city: 'Delhi'
  }
```

6. GREATER THAN — $gt

MongoDB Query:

- *db.students.find({marks: {$gt: 60}})*

Explanation: Finds students whose marks are greater than 60.

```
> db.students.find({marks: {$gt: 60}})
< {
    _id: ObjectId('6b1d0c4e7a92f3b5c8e1a000'),
    roll: 1,
    name: 'Aanya Gupta',
    age: 19,
    marks: 88,
    city: 'Lucknow'
  }
  {
    _id: ObjectId('6b1d0c4e7a92f3b5c8e1a001'),
    roll: 2,
    name: 'Riya Sharma',
    age: 18,
    marks: 76,
    city: 'Delhi'
  }
```

7. LESS THAN — $lt

MongoDB Query:

- *db.students.find({marks: {$lt: 60}})*

```
> db.students.find({marks: {$lt: 60}})
< {
    _id: ObjectId('6b1d0c4e7a92f3b5c8e1a008'),
    roll: 9,
    name: 'Sneha Rao',
    age: 18,
    marks: 58,
    city: 'Kanpur'
  }
  {
    _id: ObjectId('6b1d0c4e7a92f3b5c8e1a009'),
    roll: 10,
    name: 'Vivek Joshi',
    age: 20,
    marks: 55,
    city: 'Delhi'
  }
AcademicDB >
```

8. GREATER THAN OR EQUAL TO — $gte

MongoDB Query:

- *db.students.find({marks: {$gte: 90}})*

```
> db.students.find({marks: {$gte: 90}})
< {
    _id: ObjectId('6b1d0c4e7a92f3b5c8e1a002'),
    roll: 3,
    name: 'Rahul Verma',
    age: 19,
    marks: 92,
    city: 'Kanpur'
  }
  {
    _id: ObjectId('6b1d0c4e7a92f3b5c8e1a007'),
    roll: 8,
    name: 'Arjun Nair',
    age: 20,
    marks: 90,
    city: 'Delhi'
  }
```

9. LESS THAN OR EQUAL TO — $lte

MongoDB Query:

db.students.find({marks: {$lte: 70}})

```
> db.students.find({marks: {$lte: 70}})
< {
    _id: ObjectId('6b1d0c4e7a92f3b5c8e1a003'),
    roll: 4,
    name: 'Aman Singh',
    age: 18,
    marks: 69,
    city: 'Delhi'
  }
  {
    _id: ObjectId('6b1d0c4e7a92f3b5c8e1a008'),
    roll: 9,
    name: 'Sneha Rao',
    age: 18,
    marks: 58,
    city: 'Kanpur'
  }
  {
    _id: ObjectId('6b1d0c4e7a92f3b5c8e1a009'),
    roll: 10,
    name: 'Vivek Joshi',
    age: 20,
    marks: 55,
    city: 'Delhi'
  }
```

10. UPDATE OPERATION

MongoDB Query:

- *db.students.updateOne(*
  {roll:4},
  {$set:{marks:72}}
)

```
> db.students.updateOne({roll: 4}, {$set: {marks: 72}})
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1,
    modifiedCount: 1,
    upsertedCount: 0
  }
AcademicDB >
```

```
> db.students.find({roll: 4})
< {
    _id: ObjectId('6b1d0c4e7a92f3b5c8e1a003'),
    roll: 4,
    name: 'Aman Singh',
    age: 18,
    marks: 72,
    city: 'Delhi'
  }
AcademicDB >
```

11. DELETE ONE OPERATION

MongoDB Query:

- *db.students.deleteOne({roll:10})*

```
> db.students.deleteOne({roll: 10})
< {
    acknowledged: true,
    deletedCount: 1
  }
AcademicDB >
```

12. DELETE MANY OPERATION

MongoDB Query:

```
db.students.deleteMany({
  marks: {$lt:60}
})
```

```
> db.students.deleteMany({marks: {$lt: 60}})
< {
    acknowledged: true,
    deletedCount: 1
  }
AcademicDB >
```

13. AND OPERATION — $and

MongoDB Query:

- *db. students. find({*
  $and:[
  {marks:{$gt:80}},
  {age:21}
  ]
  })

```
> db.students.find({ $and: [ {marks: {$gt: 80}}, {age: 21} ] })
<
AcademicDB >
```

14. OR OPERATION — $or

MongoDB Query:

- *db.students.find({*
  $or:[
  {city:"Lucknow"},
  {city:"Kanpur"}]})

```
> db.students.find({ $or: [ {city: "Lucknow"}, {city: "Kanpur"} ] })
< {
    _id: ObjectId('6b1d0c4e7a92f3b5c8e1a000'),
    roll: 1,
    name: 'Aanya Gupta',
    age: 19,
    marks: 88,
    city: 'Lucknow'
  }
  {
    _id: ObjectId('6b1d0c4e7a92f3b5c8e1a002'),
    roll: 3,
    name: 'Rahul Verma',
    age: 19,
    marks: 92,
    city: 'Kanpur'
  }
```

15. SORT OPERATION

MongoDB Ascending order Query:

- *db.students.find().sort({marks : 1})*

```
> db.students.find().sort({marks: 1})
< {
    _id: ObjectId('6b1d0c4e7a92f3b5c8e1a003'),
    roll: 4,
    name: 'Aman Singh',
    age: 18,
    marks: 72,
    city: 'Delhi'
  }
  {
    _id: ObjectId('6b1d0c4e7a92f3b5c8e1a006'),
    roll: 7,
    name: 'Pooja Mishra',
    age: 19,
    marks: 73,
    city: 'Lucknow'
  }
  {
    _id: ObjectId('6b1d0c4e7a92f3b5c8e1a001'),
    roll: 2,
    name: 'Riya Sharma',
    age: 18,
    marks: 76,
    city: 'Delhi'
  }
```


By Aanya Gupta

B. MongoDB Descending Order Query:

- *db.students.find().sort({marks: -1})*

```
> db.students.find().sort({marks: -1})
< {
    _id: ObjectId('6b1d0c4e7a92f3b5c8e1a002'),
    roll: 3,
    name: 'Rahul Verma',
    age: 19,
    marks: 92,
    city: 'Kanpur'
  }
  {
    _id: ObjectId('6b1d0c4e7a92f3b5c8e1a007'),
    roll: 8,
    name: 'Arjun Nair',
    age: 20,
    marks: 90,
    city: 'Delhi'
  }
  {
    _id: ObjectId('6b1d0c4e7a92f3b5c8e1a000'),
    roll: 1,
    name: 'Aanya Gupta',
    age: 19,
    marks: 88,
    city: 'Lucknow'
  }
```

16. LIMIT OPERATION

MongoDB Query:

- *db.students.find().limit(5)*

```
> db.students.find().limit(5)
< {
    _id: ObjectId('6b1d0c4e7a92f3b5c8e1a000'),
    roll: 1,
    name: 'Aanya Gupta',
    age: 19,
    marks: 88,
    city: 'Lucknow'
  }
  {
    _id: ObjectId('6b1d0c4e7a92f3b5c8e1a001'),
    roll: 2,
    name: 'Riya Sharma',
    age: 18,
    marks: 76,
    city: 'Delhi'
  }
  {
    _id: ObjectId('6b1d0c4e7a92f3b5c8e1a002'),
    roll: 3,
    name: 'Rahul Verma',
    age: 19,
    marks: 92,
    city: 'Kanpur'
  }
  {
    _id: ObjectId('6b1d0c4e7a92f3b5c8e1a003'),
    roll: 4,
    name: 'Aman Singh',
    age: 18,
    marks: 72,
    city: 'Delhi'
  }
  {
    _id: ObjectId('6b1d0c4e7a92f3b5c8e1a004'),
    roll: 5,
    name: 'Neha Yadav',
    age: 19,
    marks: 81,
    city: 'Lucknow'
  }
```


By Aanya Gupta

17. SEARCH BY CITY

MongoDB Query:

- *db.students.find({city:"Lucknow"})*

```
> db.students.find({city:"Lucknow"})
< {
    _id: ObjectId('6b1d0c4e7a92f3b5c8e1a000'),
    roll: 1,
    name: 'Aanya Gupta',
    age: 19,
    marks: 88,
    city: 'Lucknow'
  }
  {
    _id: ObjectId('6b1d0c4e7a92f3b5c8e1a004'),
    roll: 5,
    name: 'Neha Yadav',
    age: 19,
    marks: 81,
    city: 'Lucknow'
  }
  {
    _id: ObjectId('6b1d0c4e7a92f3b5c8e1a006'),
    roll: 7,
    name: 'Pooja Mishra',
    age: 19,
    marks: 73,
    city: 'Lucknow'
  }
```

18. COUNT DOCUMENTS

MongoDB Query:

- *db.students.countDocuments()*

```
> db.students.countDocuments()
< 8
```

19.UPDATE MANY OPERATION

MongoDB Query:

- *db.students.updateMany({city: "Lucknow"}, {$set: {state: "UP"}})*

```
> db.students.updateMany({city: "Lucknow"}, {$set: {state: "UP"}})
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 3,
    modifiedCount: 3,
    upsertedCount: 0
  }
```

```
> db.students.findOne({roll: 1})
< {
    _id: ObjectId('6b1d0c4e7a92f3b5c8e1a000'),
    roll: 1,
    name: 'Aanya Gupta',
    age: 19,
    marks: 88,
    city: 'Lucknow',
    state: 'UP'
  }
```

20.DISTINCT

MongoDB Query:

- *db.students.distinct("city")*

```
> db.students.distinct("city")
< [
    'Delhi',    'Kanpur',
    'Lucknow'
  ]
```